

# Combat System — Inkglass Odyssey

*Design document. Read before working on relay/combat/, relay/checks/, or any combat-related endpoint.*

*Cross-references: docs/economy\_balance.md (HP tables, damage scaling), docs/companion\_system.md (companion combat), docs/schemas\_reference.md (ability.json, character\_sheet.json).*

## Specialisation Paths and Hit Points

Each world defines specialisation paths (class equivalents) in world\_config.json. Every path carries a hit\_die size, two saving\_throw\_proficiencies, primary\_ability\_scores, and available\_skill\_proficiencies. The path determines HP calculation and narrative framing but does not restrict ability or item access — it shapes the character, not the game.

HP formula: Level 1 HP = hit die maximum + CON modifier. Each subsequent level adds hit die average (rounded up) + CON modifier.

| Path archetype    | Hit die | Lv1 HP (CON 14) | Lv5 HP | Lv10 HP | Lv20 HP |
|-------------------|---------|-----------------|--------|---------|---------|
| Scholar / caster  | d6      | 8               | 28     | 53      | 103     |
| Balanced / rogue  | d8      | 10              | 34     | 64      | 124     |
| Martial / fighter | d10     | 12              | 40     | 75      | 145     |
| Tank / guardian   | d12     | 14              | 46     | 86      | 166     |

These HP values are the reference anchor for all damage, healing, and pricing balance. When authoring weapons, abilities, consumables, or encounter difficulty, scale against these numbers.

NPC HP uses the same formula. The Workshop calculates and displays NPC max HP automatically from their level, hit die, and CON score — authors do not compute it manually.

## Armour Class (AC)

AC = base 10 (unarmoured) + armour bonus + DEX modifier (capped by armour category) + shield bonus + magical modifiers.

| Armour category   | DEX cap          |
|-------------------|------------------|
| None (unarmoured) | No cap           |
| Light             | No cap           |
| Medium            | +2 maximum       |
| Heavy             | +0 (DEX ignored) |

The relay computes AC from equipped gear and ability scores. The LLM never sees AC values — it receives narrative descriptions of armour. The relay resolves hit/miss.

## Attack Resolution

Step 1 — Attack roll:  $d20 + \text{ability modifier (per weapon's modifier\_source)} + \text{proficiency bonus (if proficient with weapon type)}$ . Natural 20: critical hit. Natural 1: automatic miss.

Step 2 — Compare to target's AC. Hit if roll meets or exceeds AC.

Step 3 — Damage roll:  $\text{weapon damage\_dice} + \text{ability modifier}$ . Critical hits double the number of dice (not the modifier). So  $2d6+3$  becomes  $4d6+3$  on a crit.

## Saving Throws

When a target is affected by something that isn't a direct attack (spells, traps, poisons, mind control), the defender rolls a saving throw:  $d20 + \text{ability modifier} + \text{proficiency bonus (if proficient in that save)}$ . Each specialisation path grants proficiency in two saving throw categories.

Save DC (set by the attacker):  $8 + \text{attacker's ability modifier} + \text{attacker's proficiency bonus}$ .

Abilities that force saves carry a `save_type` (which ability the defender uses) and `save_dc_source` (which ability the attacker uses to set the DC) in their schema.

## Advantage and Disadvantage

Situational modifiers use D&D 5e's advantage/disadvantage system rather than stacking numerical bonuses. Roll 2d20, take the higher (advantage) or lower (disadvantage).

Mathematically equivalent to roughly  $\pm 3.5$ .

The implicit check system flags advantage/disadvantage from scene state: attacking from ambush (advantage), frightened and making a WIS save (disadvantage), attacking from high

ground at range (advantage). The LLM proposes advantage/disadvantage in its first-call analysis; the relay validates against scene state and active conditions.

Advantage and disadvantage cancel each other out — any number of each reduces to: both present = straight roll.

Do not stack numerical situational modifiers. Use advantage/disadvantage only.

## Damage Types and Resistances

Damage types: slashing, piercing, bludgeoning (physical); fire, cold, lightning, thunder, poison, acid, necrotic, radiant, force, psychic (magical/elemental).

| Defense       | Effect                     |
|---------------|----------------------------|
| Resistance    | Half damage of that type   |
| Vulnerability | Double damage of that type |
| Immunity      | Zero damage of that type   |

NPCs and fauna carry resistances, vulnerabilities, and immunities arrays in their stat blocks. The relay applies multipliers after the damage roll — never pre-roll. Most creatures should have at most 1–2 resistances and 1 vulnerability. Immunities should be rare and thematically justified.

## Conditions and Status Effects

Active conditions are tracked on character sheets and NPC stat blocks as an array with duration or expiry turn. Every condition entry must specify when it ends — no open-ended conditions.

| Condition  | Mechanical effect                                                            |
|------------|------------------------------------------------------------------------------|
| Poisoned   | Disadvantage on attack rolls and ability checks                              |
| Frightened | Disadvantage on checks while source is visible; cannot move closer to source |
| Charmed    | Cannot attack the charmer; charmer has advantage on social checks            |
| Stunned    | Cannot move or act; auto-fail STR and DEX saves                              |
| Restrained |                                                                              |

|                            |                                                                                                                                        |
|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
|                            | Speed 0; disadvantage on DEX saves; attacks against have advantage                                                                     |
| Incapacitated              | Cannot take actions; others have advantage on attacks                                                                                  |
| Exhaustion<br>(tiered 1–6) | Cumulative: 1=disadvantage on checks, 2=speed halved, 3=disadvantage on attacks/saves, 4=HP max halved, 5=speed 0, 6=character retired |

Exhaustion is the key non-combat condition. Caused by: forced marches, starvation, magical drain, extended injury, death state recovery. Creates mechanical pressure from narrative situations.

## Action Economy

Each combat turn allows: one major action (attack, use ability, interact mechanically), one minor action (move, draw weapon, speak briefly), and one reaction (opportunity attack, counter — triggered on another's turn).

If the player's prose describes more than one major action, the relay resolves only the first and narrates the remainder as interrupted or deferred. This is documented in docs/prompt\_engineering.md so the LLM paces its prose responses accordingly.

## Initiative

At combat start: d20 + DEX modifier for each participant. Highest goes first. In prose RP combat, the player always acts first (they wrote the prose), NPC response follows. For multi-participant combat, initiative determines NPC response order.

## Narrative Death State

At 0 HP, the character enters a compromised state (Broken, Incapacitated, or Critically Wounded per world terminology). Mechanical effects: disadvantage on all checks, speed halved, gain 1 exhaustion level per turn at 0 HP (maximum 3 from this source). Any healing above 0 HP ends the death state but does not remove accumulated exhaustion — that requires rest (the "recovery arc"). At exhaustion level 6, the character is permanently retired.

This is a ticking clock, not a coin flip. Allies are motivated to heal quickly. The recovery arc (shedding exhaustion over subsequent rests) has mechanical teeth.

## Passive Checks

Passive check value: 10 + ability modifier + proficiency bonus (if proficient). Maintained for Perception, Insight, and Investigation equivalents per character.

When scene state includes hidden elements (traps, concealed NPCs, secret doors), the relay compares the element’s DC against the character’s passive check automatically. If met, a hint is injected into the scene narrator’s prompt without the player asking to search. Creates natural discovery moments without meta-gaming.

## Rest and Recovery

| Rest type                    | Healing                                                       | Resources                                             | Availability                     |
|------------------------------|---------------------------------------------------------------|-------------------------------------------------------|----------------------------------|
| Short rest (1 hour in-world) | 25% of max HP + hit die rolls (dice = half level, rounded up) | Partial resource recovery per world_config            | Between encounters               |
| Long rest (8 hours in-world) | 100% of max HP                                                | All resources restored, exhaustion reduced by 1 level | Safe locations, between sessions |

Defined per world in world\_config.json under rest\_rules.

## Environmental Interaction

Scene state carries an environmental\_effects array defining conditions that affect checks and combat in the current scene.

| Effect                  | Mechanical impact                                          | Narrative use                 |
|-------------------------|------------------------------------------------------------|-------------------------------|
| Darkness                | Disadvantage on Perception and ranged attacks              | Horror, stealth scenarios     |
| Difficult terrain       | Movement costs doubled, disadvantage on DEX checks         | Swamps, rubble, dense forest  |
| Elevation (high ground) | Advantage on ranged attacks                                | Tactical positioning in prose |
| Extreme weather         | Disadvantage on Perception, possible exhaustion gain       | Travel pressure, wilderness   |
| Hazardous surface       | Damage per turn in area (type and dice defined per effect) | Combat terrain, trap effects  |

Environmental effects are defined per-region or per-scene in scenario files and applied by the relay when resolving checks. The LLM receives them as scene state context. The EAS may adjust movement animations (cautious in darkness, slowed in difficult terrain). No tile-based simulation required — narrative-applicable modifiers only.